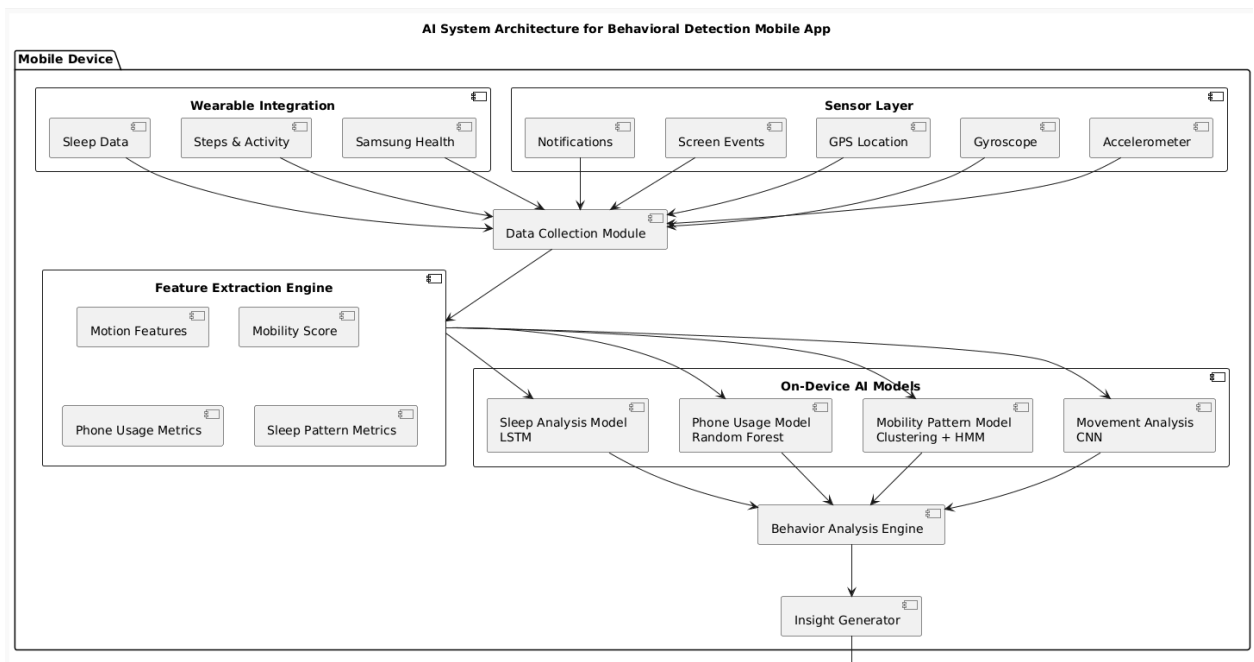


1. AI Use Cases for Behavioral Detection in a Mobile Application

The application gathers information from the following sources:

- Accelerometer
- Gyroscope
- GPS location
- Screen on/off events
- Notification interactions
- Step count and activity data from Samsung Health
- Wearable activity and sleep metrics



2. AI Use Cases in the Mobile Application

2.1 Neuro-Motor Health and Stress Detection

Sensors Used

- Accelerometer
- Gyroscope
- Wearable motion data

Scenario

The mobile application monitors movement patterns while the user carries the phone or wears a smartwatch.

Examples:

- Hand tremors while holding the phone
- Changes in walking rhythm when the phone is in a pocket
- Increased restlessness or reduced physical stability

AI Advantage

Traditional health applications typically focus only on **step counting or activity tracking**. AI models can analyze motion signals to detect subtle movement patterns that may indicate:

- Stress
- Physical fatigue
- Reduced mobility

2.2 Social Withdrawal Detection (Mobility Analysis)

Sensors Used

- GPS location
- Location clusters

Scenario

The mobile application analyzes the user's daily mobility patterns including the locations visited and the geographical area covered throughout the day.

example:

- Normal routine: Home → Work → Gym → Social places
- Reduced mobility: Staying mostly at home or visiting fewer places

AI Advantage

A significant reduction in mobility diversity or travel distance may indicate social withdrawal.

AI can identify:

- Reduced movement area
 - Fewer visited locations
 - Deviations from daily routines
-

2.3 Compulsive Phone Usage Detection

Sensors Used

- Screen on/off events
- Notification timing
- App usage categories

Scenario

The application analyzes how users interact with their phones throughout the day.

Examples include:

- Unlocking the phone immediately after receiving a notification
- Frequent phone checks without notifications
- Many short phone sessions during the day

AI Advantage

Most digital wellbeing apps measure only total screen time. However, AI can detect behavioral patterns associated with anxiety or compulsive device checking, such as:

- High-frequency phone interactions
 - Notification-driven behavior
 - Repeated short usage sessions
-

2.4 Sleep Behavior and daily Rhythm Monitoring

Sensors Used

- Wearable sleep data
- Phone usage patterns
- Screen on/off events

Scenario

The mobile application combines wearable sleep metrics with smartphone activity data to better understand the user’s sleep habits and circadian rhythm.

Examples:

- Late-night phone usage before sleep
- Frequent night time phone interactions
- Inconsistent sleep schedules

AI Advantage

Wearable devices sometimes incorrectly detect sleep when the user is simply lying still. By combining wearable sleep information with smartphone usage data, AI models can provide more accurate sleep behavior analysis.

AI Use Case	Sensors	Behavioral Indicators	ML Technique	Impact
Neuro-Motor Monitoring	Accelerometer, Gyroscope	tremors, unstable motion patterns	Random Forest	Detect fatigue or stress
Mobility & Social Activity	GPS	reduced movement area, routine changes	Random Forest	Detect social withdrawal
Compulsive Phone Usage	Screen events, notifications	frequent checking, short sessions	Random Forest	Detect anxiety-related usage
Sleep Behavior Monitoring	Wearable sleep data, phone usage	irregular sleep patterns	Random Forest	Sleep quality insights

AI Use Case	Sensors	Behavioral Indicators	ML Technique	Impact
Routine Deviation Detection	GPS, steps, activity data	abnormal daily routine	Random Forest	Detect unusual behavior